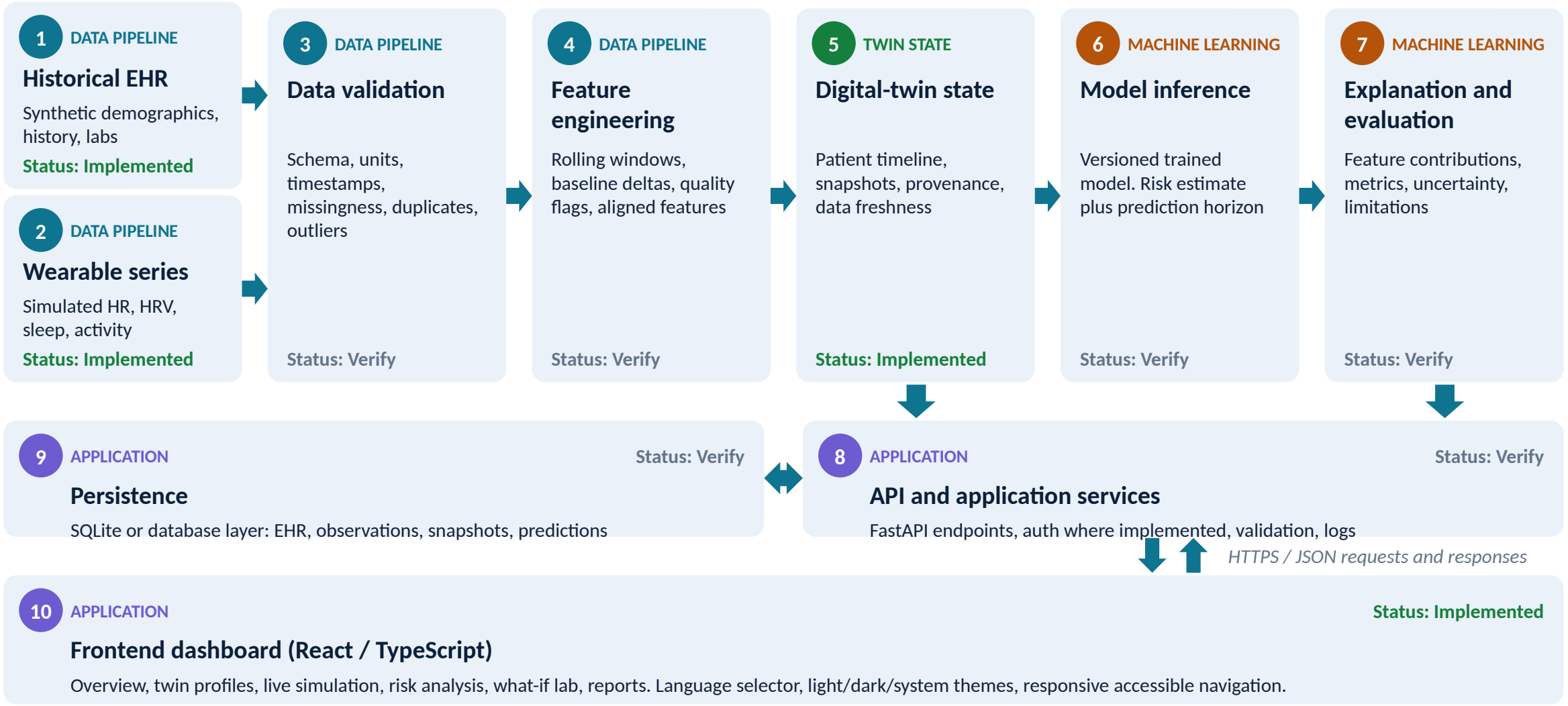
End-to-End System Architecture



Component Responsibilities

Component	Responsibility	Input to output	Status
Synthetic EHR	Simulated demographics, history, conditions, labs	Patient record to historical context	IMPLEMENTED
Wearable simulator	Timestamped HR, HRV, sleep, activity	Settings to time series	IMPLEMENTED
Validation and features	Unit, timestamp, missingness checks; build model features	Raw records to feature vector	VERIFY
Digital-twin state	Combine history with recent features; keep snapshots	EHR + features to snapshot	IMPLEMENTED
ML inference and evaluation	Versioned model, horizon, validation metrics	Features to estimate and metrics	VERIFY
API and persistence	FastAPI services; store records, snapshots, predictions	Frontend and data, both ways	VERIFY
Frontend dashboard	Trends, model outputs, scenarios, reports	API responses to interactive UI	IMPLEMENTED

Statuses come from the live prototype where visible; Verify = confirm in the repository.